

Phase 3 P0 Implementation Report

Project: Vyapar AI Employee Platform | **Tenant (pilot):** HONS | **Date:** June 1, 2026 | **Source of truth:** Recent Test Report (June 3, 2026)

Status: P0 IMPLEMENTED — READY FOR DEPLOY + SMOKE TEST

All 5 P0 modules built, wired into `ai_employee_engine.py` , and covered by unit tests. P1 (Dialogue Planner, Persona Manager, Escalation Handler) intentionally not started until P0 tests pass in production.

1. Executive Summary

Phase 3 P0 addresses the behavioral failures identified in the June 3 Telegram test session. The root cause was not lead detection — it was missing **conversation state**, **turn routing**, and **response validation**. P0 adds a session brain layer between Telegram ingress and the LLM, enforcing rules the prompt alone could not guarantee.

2. P0 Modules Delivered

#	Module	File	Function
1	Session State Manager	<code>session_state_manager.py</code>	Tracks user_id, company_id, language, name, phone, location, package_interest, lead_stage, pitch_count, phone_collected, escalation_requested per tenant session
2	Turn Router	<code>turn_router.py</code>	Routes: greeting, company_info, support, sales, objection, escalation, correction, memory_query, general_knowledge. Returns direct answers without LLM where safe
3	Response Validator	<code>response_validator.py</code>	Blocks repeated package pitch, repeated phone request, repeated closing sentence, duplicate responses (>70% overlap), hallucinated phone numbers

4	Language Lock	<code>language_lock.py</code>	Detects Nepali vs English from user text; injects lock prompt so LLM does not randomly switch language
5	Memory Validator	<code>memory_validator.py</code>	Saves name only from high-confidence patterns; rejects "I am Indian I am Nepali"; phone saved only if present in source message

3. Database Migration

New table: `conversation_state` (composite primary key: `user_id` + `company_id`)

Column	Type	Purpose
<code>user_id, company_id</code>	VARCHAR (PK)	Session identity per tenant
<code>language</code>	VARCHAR(16)	nepali english
<code>name, phone, location</code>	VARCHAR	Cached customer facts
<code>package_interest, lead_stage</code>	VARCHAR	Sales context
<code>pitch_count</code>	INTEGER	Times package was pitched
<code>phone_collected</code>	BOOLEAN	Stops re-asking phone
<code>escalation_requested</code>	BOOLEAN	Escalation turn tracking
<code>last_assistant_reply</code>	TEXT	Duplicate detection baseline
<code>turn_count</code>	INTEGER	Session turn counter

Auto-created on startup via `init_db()` → SQLAlchemy `create_all` .

Manual SQL for existing Postgres: `docs/migrations/001_conversation_state.sql`

4. Pipeline Architecture (After P0)

```
Telegram update
  → main.py (update_id dedup – prevents double replies)
  → sanitize_user_text
  → detect_language → session state sync
  → extract_memory_facts → memory_validator → save memory
  → turn_router
    ├── direct_answer? → response_validator → reply (no LLM)
    └── memory_query? → direct memory answer
```

- lead extraction + upsert → session sync
- sales mode (turn-aware, not always ON)
- compose_system_prompt (+ language lock + session rules + turn hint)
- OpenAI
- response_validator (strip/blocks repeat pitch, phone, closing)
- record session state + save chat turn
- Telegram reply

5. Success Criteria — Before vs After

Criterion	Before (June 3 test)	After (P0)	Status
Phone given once → never asked again	Re-asked every turn	<code>phone_collected=True</code> + validator strips phone asks	Implemented
"Kun net ho?" → company name only	100 Mbps package pitch	Direct answer: "Yo Himalayan Online Service Pvt. Ltd. ko official AI employee ho."	Implemented
"Tapaiko naam ke ho?" → AI identity	Awkward "pratinidhi" reply	Direct: "Ma ... ko AI employee hu..."	Implemented
Nepali session stays Nepali	English/Nepali mix	<code>language_lock_prompt</code> injected into system prompt	Implemented
No repeated package pitch	Same 100 Mbps pitch 15+ times	Overlap $\geq 70\%$ → fallback; pitch_count rules in prompt	Implemented
No hallucinated phone	Invented 9898989898	Phone saved only if in user text; validator strips unknown numbers	Implemented
No duplicate Telegram replies	Double messages sent	<code>update_id</code> dedup cache in main.py	Implemented

6. Changed Files

File	Change
<code>session_state_manager.py</code>	NEW — session CRUD + prompt block
<code>turn_router.py</code>	NEW — 9 turn routes + direct answers

<code>response_validator.py</code>	NEW — post-LLM validation
<code>language_lock.py</code>	NEW — language detection + lock
<code>memory_validator.py</code>	NEW — high-confidence fact filtering
<code>memory_db.py</code>	Added ConversationState model
<code>ai_employee_engine.py</code>	Full P0 pipeline integration
<code>prompts.py</code>	Session state, language lock, turn hint blocks
<code>main.py</code>	Telegram update_id deduplication
<code>tests/test_p0_phase3.py</code>	NEW — 17 P0 unit tests
<code>tests/test_sales_mode.py</code>	Updated for turn_route parameter
<code>docs/migrations/001_conversation_state.sql</code>	NEW — manual migration reference

7. Test Coverage

Run locally:

```
cd "Z:\Vyapar AI\cursor"
python -m unittest discover -s tests -v
```

Test file	Covers
<code>test_p0_phase3.py</code>	Memory validator, language lock, turn router, session state, response validator, success criteria
<code>test_sales_mode.py</code>	Sales mode activation with turn_route
<code>test_sales_objection.py</code>	Objection detection (Phase 2)
<code>test_lead_phase2.py</code>	Lead scoring/staging (Phase 2)

8. Deploy Checklist

1. Deploy **all files atomically** to Render (partial deploy causes import errors)
2. Restart service → `init_db()` creates `conversation_state` table
3. Run unit tests on deploy machine before going live
4. Telegram smoke test — send these 7 messages in order (Nepali):

#	Message	Expected
1	Namaste	Warm greeting, no package pitch
2	Kun net ho?	Company name only, no Mbps pitch
3	Tapaiko naam ke ho?	AI employee identity, not user name
4	Mero naam Ram ho	Acknowledge name saved
5	9801234567	Phone saved, not re-asked later
6	100 Mbps kati ho?	Price answer, one pitch only
7	(repeat question 6)	No repeated full package pitch

9. P1 — Not Started (By Design)

Module	Status
6. Dialogue Planner	Pending — after P0 smoke test passes
7. Persona Manager	Pending
8. Escalation Handler	Pending
9. Telegram Inline Keyboard	P2
10. Meta Conversation Handler	P2

10. Recommended Git Commits

```
git add memory_db.py session_state_manager.py docs/migrations/001_conversation_state.sql
git commit -m "P0: add session state manager and conversation_state table"

git add memory_validator.py
git commit -m "P0: add memory validator for high-confidence name and phone"

git add language_lock.py
git commit -m "P0: add language lock for Nepali/English sessions"

git add turn_router.py
git commit -m "P0: add turn router with direct answers for info and identity"

git add response_validator.py
```

```
git commit -m "P0: add response validator to block repeat pitch and phone asks"
```

```
git add ai_employee_engine.py prompts.py main.py tests/
```

```
git commit -m "P0: wire session state, router, validators into employee engine"
```

Vyapar AI Employee Platform — Phase 3 P0 Implementation Report — Generated June 1, 2026

Workspace: Z:\Vyapar AI\cursor | Pilot tenant: HONS (company_profiles.json)